Diagnosing Mitigation Need: Controlled Shortages and Signal-Matched Interventions in Histopathology Patch Classification

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Abstract

A class can be underserved for several distinct reasons, and the count that marks it as rare does not say which. In computational pathology, raw sample counts hide at least one further shortage — few labelled patients or slides behind many informative patches — and say nothing about how separable a class is under the available representation or how much variation it spans. This benchmark uses training allocation as a controlled way to induce such a shortage, and then asks which signal a mitigation method should read to repair it: how would the same prediction task change if one fixed training budget were distributed evenly across classes rather than concentrated in head classes, which signal profile does the resulting condition actually present, and does a method that reads the signal matching that profile recover more of the damage than one that reads a different signal? CAMELYON16, TCGA-UT, BRACS, and PANDA are studied in patch classification using frozen Virchow2 features. Patient- or case-disjoint partitions, natural validation and test cohorts, model family, and optimizer-update budgets remain fixed while only training support is redistributed among size-matched balanced, moderate, and severe conditions. Balanced accuracy and tail-group macro negative log likelihood measure discrimination and probability quality. Mitigation recovery is interpreted only after a prespecified gate establishes a meaningful allocation-induced deficit. A five-member family of methods — one per signal, all a mean-one class weight on unmodified cross-entropy — makes the signal contrast interpretable at a fixed intervention level. Exploratory analyses then relate damage and recovery to achieved allocation severity, independent-evidence shortage, support–difficulty alignment, and representational-diversity shortage. The benchmark is scoped to patch classification, where each prediction unit carries its own label.

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1 Introduction

Choosing a mitigation method requires first identifying what limits a class. In histopathology patch classification, an underserved class may lack nominal patch support, independent evidence from patients or slides, learnable signal, or representational diversity. These shortages are not interchangeable. Thousands of patches from a few patients may provide less independent evidence than a smaller patch set drawn from many patients, while neither count reveals how difficult a class is to separate or how much variation it contains. A method that responds to one shortage may therefore be ineffective when another shortage limits performance.

This benchmark asks when such a shortage causes enough damage to warrant mitigation and whether recovery improves when a method reads the signal that identifies that shortage. Training allocation provides the controlled intervention needed to answer these questions. For the same task and total training budget, examples are distributed either evenly across classes or concentrated in head classes. A size-matched balanced cross-entropy (CE) model provides the reference. Validation and test cohorts retain their natural class distributions, so the resulting contrast reflects the training allocation rather than a change in the evaluation population.

Allocation is thus a way to create controlled variation in evidence, not the benchmark's main explanatory focus. Each condition is instead characterized by its realized *signal profile*: class prevalence, nominal patch support, independent patient or slide support, class difficulty, and representational diversity. This profile is measured rather than inferred from the nominal imbalance ratio. The companion methods report defines the signal taxonomy and locates each tested method within it (Queisler, 2026b); the present benchmark tests whether matching method to signal matters empirically.

The study covers CAMELYON16, TCGA-UT, BRACS, and PANDA using frozen Virchow2 patch features. Patient- or case-disjoint partitions, model family, optimizer-update budget, and natural validation and test cohorts remain fixed while the allocation of a common training budget changes. Balanced, moderate, and severe allocations establish the controlled comparisons. To separate signal choice from intervention mechanism, the confirmatory family contains five methods that all apply a mean-one class weight to unmodified CE. Their weights differ only in whether they use prevalence, nominal support, independent support, difficulty, or diversity.

This design yields three research questions:

RQ1. Which shortage does controlled allocation create, and what does it damage? Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

RQ1 first establishes whether a recoverable deficit exists and which signals characterize it. RQ2 is evaluated only after this deficit passes a prespecified gate. RQ3 then examines why deficit size and recovery headroom vary; it is an exploratory association analysis, not another method ranking.

The benchmark is limited to patch classification because directly labelled patches allow nominal patch support to change within a fixed pool of patients or slides. At whole-slide level, reallocating labelled slides changes nominal and independent support together, making the two signals harder to separate. Dataset provenance and detailed method objectives remain in the companion reports (Queisler, 2026a,b). The next section defines the prediction and evidence units on which the experimental design rests.

2 Study Setting

This benchmark covers one prediction regime: patch classification predicts a label for one directly labelled image crop, so every training example carries its own target. The terms *patch* and *tile* both denote a 256×256 crop, but *patch* is used throughout.

Table 1 gives the patch target and patient/case grouping used by each dataset.

Dataset	Patch-level example and target	Patient/case grouping
CAMELYON16	Non-overlapping tissue crop at 20x; tumor if at least 50% of its aligned mask cell is tumor, normal otherwise	Slide (one slide per patient)
TCGA-UT	Author-supplied tumor-region crop; inherits its source WSI's cancer type	TCGA participant
BRACS	Non-overlapping crop from a pathologist-annotated ROI; inherits one of seven ROI subtypes	Patient
PANDA	Foreground crop; cancer if at least 50% of its aligned provider-specific mask cell is cancer, benign otherwise	Biopsy (one WSI)

Table 1: Dataset-specific definitions of patch-level examples. TCGA-UT's eligible target set omits one of the 32 annotated cancer types; Appendix A states which class and why.

With the prediction examples, targets, and patient/case groupings established, the next section follows the controlled design from eligible examples to controlled model comparisons.

3 Controlled Experimental Design

Figure 1 gives the complete design before its steps are considered separately. Eligible examples are divided into training, validation, and test partitions by partitioning unit; training support is then formed into a natural anchor and three controlled conditions; controlled model comparisons are finally evaluated on unchanged natural validation and test cohorts. Only class allocation varies among the controlled conditions.

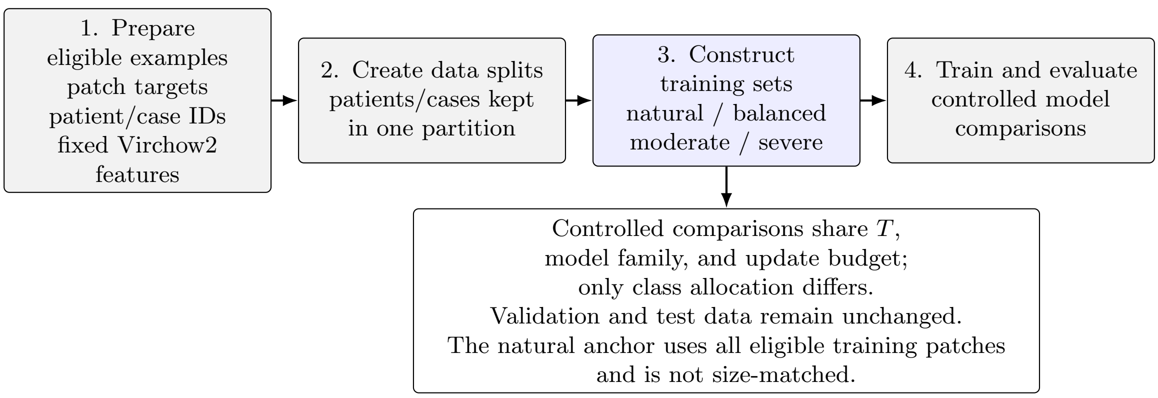

Figure 1: Controlled benchmark design. Gray boxes define shared evidence, partitioning, and evaluation; the blue box contains the training-support intervention.

3.1 Eligible examples and data partitions

A patch becomes an *eligible example* only if it has a valid target and patient/case identifier, satisfies its dataset's tissue and annotation rules, and has readable fixed features. Eligibility is determined once, before partitions or training conditions are constructed. Every eligible patch is embedded once with a frozen Virchow2 encoder; concatenating the CLS token and mean patch token gives 2560 input features (Vorontsov et al., 2024; Zimmermann et al., 2024). Appendix A gives the complete exclusion, preprocessing, and provenance rules.

Within each data split, a *partition* is one of three mutually exclusive subsets: training, validation, or test. A *partitioning unit* is the patient or participant where available and the biopsy otherwise. Exactly three partitioning-unit-disjoint data splits are derived from the eligible examples, and every patch from one unit follows the same assignment. Imbalance manipulation is confined to the training partition, so every condition retains the same natural validation and test partitions. No held-out example is deleted, duplicated, or reweighted according to training severity. Validation selects hyperparameters and checkpoints; test evaluation uses every eligible patch at one fixed prevalence.

3.2 Training-support construction

Four training conditions separate two purposes. The *natural anchor* uses all eligible training patches and describes performance at the dataset's observed prevalence. The other three conditions form the controlled experiment: a balanced reference, a moderate imbalance targeting $\rho = 10$, and a severe imbalance targeting $\rho = 100$. These controlled conditions contain the same total number of training patches T; only their allocation among classes differs. They carry the full method roster and enter the imbalance-deficit and recovery estimands, whereas the natural anchor carries the CE reference only.

The controlled comparison asks how the same prediction task changes when a fixed training budget is distributed evenly across classes or concentrated in head classes. Figure 2 makes this intervention concrete using one BRACS split. The natural allocation shows the source distribution, but it is not size-matched to the three controlled allocations.

The example shows why the natural anchor and controlled conditions must not be conflated. Natural BRACS is already imbalanced ($\rho_{\text{achieved}} = 5.90$) and contains 217,399 training patches, almost 13 times the controlled budget. The controlled allocations instead hold $T = 16{,}796$ fixed and redistribute it: their achieved ratios are 1.00, 10.01, and 100.47. Other tail assignments map the same moderate and severe support profiles to different semantic classes; they do not change T or the severity profile itself.

The shared T is selected jointly with severity rather than from size alone. Construction retains the largest feasible total for which every tail assignment realizes moderate $\rho_{\text{achieved}} \in [9, 11]$ and severe $\rho_{\text{achieved}} \in [90, 110]$. Appendix A specifies the search and feasibility rules.

Within each controlled allocation, severity determines how the fixed total T is distributed across support ranks. Binary tasks have one head and one tail. Multiclass tasks place intermediate supports at equal intervals on a logarithmic scale, as the BRACS profiles in Figure 2 illustrate. A moderate or severe construction that collapses to the balanced allocation is marked degenerate and is not pooled with genuine imbalanced conditions.

The target ratio $\rho_{\text{target}} \in \{1, 10, 100\}$ specifies the intended head–tail contrast. Because a training set contains whole patches and each class has a limited eligible inventory, that ratio may not be realized exactly. The achieved ratio is therefore recomputed from every training set. For realized class supports $N_1, \ldots, N_K$, it is

$$\rho_{\text{achieved}} = \frac{N_{\max}}{N_{\min}}. \tag{1}$$

This ratio records the most extreme head–tail contrast. The complementary normalized-entropy

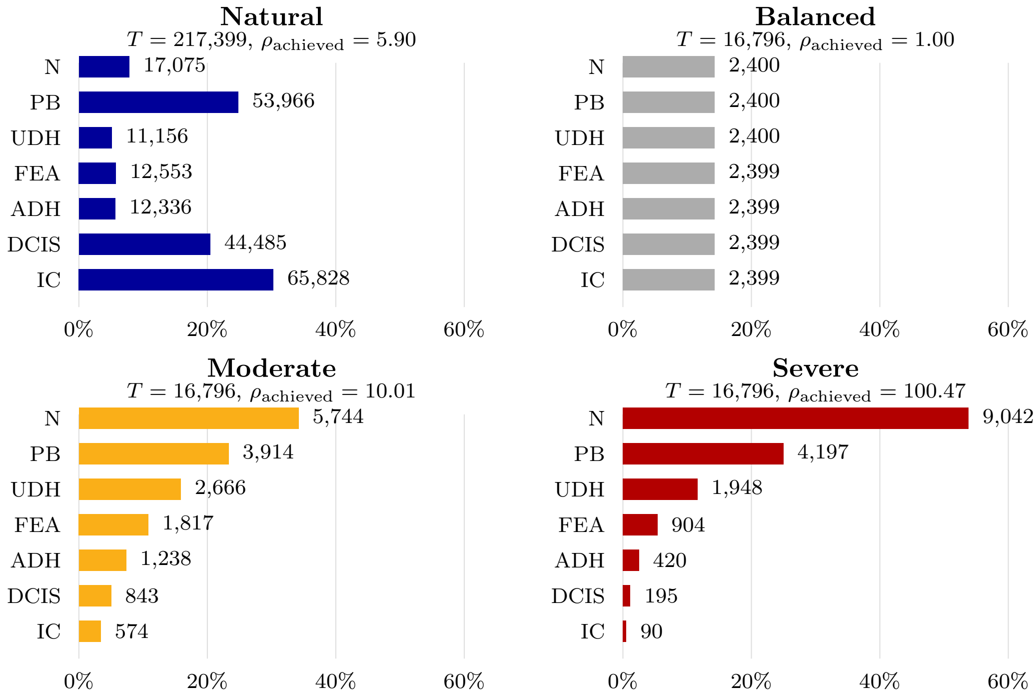

Figure 2: Realized BRACS patch allocations. Bars show each class's share of training patches; labels give realized patch counts. Moderate and severe use the native tail assignment. The natural anchor uses all eligible training patches, while balanced, moderate, and severe share $T = 16{,}796$. Class abbreviations are N (normal), PB (pathological benign), UDH (usual ductal hyperplasia), FEA (flat epithelial atypia), ADH (atypical ductal hyperplasia), DCIS (ductal carcinoma in situ), and IC (invasive carcinoma).

deficit

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{c=1}^{K} p_c \log p_c}{\log K}, \qquad p_c = \frac{N_c}{\sum_j N_j}. \tag{2}$$

uses the complete class distribution and is normalized for class count. It equals zero for a balanced allocation and increases as support becomes concentrated. Both summaries are computed from realized rather than requested counts. Table 2 separates overall training support from condition-specific imbalance. Its upper part contrasts the natural anchor with the controlled realizations; its lower part shows how their class distributions differ.

3.3 Tail assignments

A *tail assignment* determines which semantic class receives each support rank. The native assignment follows the dataset's clinical or ordinal class order. The difficulty-aligned assignment gives less support to harder classes, compounding scarcity and difficulty; the reversed assignment gives harder classes more support. Duplicated mappings are omitted transparently. Figure 3 illustrates how the same severity profile yields different semantic tails.

Detailed support floors, allocation equations, infeasibility rules, seed families, and locked controls are specified in Appendix A and Table 5. The next section asks how controlled model comparisons turn that evidence into damage and recovery estimates.

Dataset	Natural			Controlled		
	Units	Slides	Patches	Units	Slides	Patches
BRACS	103	262–273	193,010–217,704	78–89	173–233	13,982–27,202
CAMELYON16	264	264	2,139,913–2,234,906	103–118	103–118	28,810–36,127
PANDA	7,358	7,358	3,141,372–3,161,828	6,025–6,080	6,025–6,080	1,099,847–1,120,125
TCGA-UT	5,005	6,090–6,135	1,120,260–1,129,060	750–914	784–1,012	46,371–69,317

Dataset	Metric	Natural	Balanced	Moderate	Severe
BRACS	ρ_{achieved}	5.722–8.297	1.000–1.001	10.003–10.007	100.301–100.467
	$1 - H_{\text{norm}}$	0.118–0.143	0.000	0.128	0.353
CAMELYON16	ρ_{achieved}	22.453–33.608	1.000	9.998–10.001	99.913–100.088
	$1 - H_{\text{norm}}$	0.746–0.811	0.000	0.560–0.561	0.920
PANDA	ρ_{achieved}	4.645–4.735	1.000	10.000	99.996–100.003
	$1 - H_{\text{norm}}$	0.326–0.332	0.000	0.561	0.920
TCGA-UT	ρ_{achieved}	25.018–25.629	1.000–1.001	9.989–10.007	99.490–100.848
	$1 - H_{\text{norm}}$	0.080–0.082	0.000	0.060	0.177

Table 2: Overall realized training support (top) and imbalance (bottom). Units are partitioning units. Entries give the minimum and maximum across three frozen splits; moderate and severe imbalance ranges additionally span all retained tail assignments. The controlled support columns summarize balanced, moderate, and severe realizations together: their patch total is shared within a split, while the selected units and slides can vary with allocation. Counts are unique within each realization, even when one unit contributes more than one class. A single value indicates an invariant quantity.

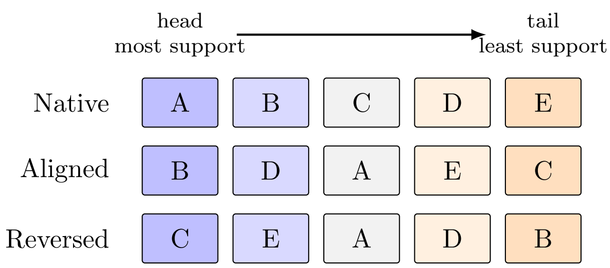

Figure 3: Illustrative tail assignments for a five-class target. Columns are support ranks fixed by the severity profile. Difficulty increases from B to C: aligned compounds difficulty with scarce support, while reversed compensates it.

4 Measuring Damage and Recovery

Section 3 established the controlled training conditions and the evidence held fixed across them. This section follows each controlled comparison from fitted models to an inferential claim. It first identifies the compared methods and their shared execution controls, then defines the endpoints used to detect imbalance-induced damage, measures mitigation recovery only where such damage is demonstrated, and finally quantifies uncertainty and tests the prespecified signal contrast. RQ1 and RQ2 are therefore answered in a deliberately asymmetric sequence: balanced CE is compared with imbalanced CE first, and only when that contrast shows a meaningful deficit on discrimination or calibration is a mitigation method interpreted in terms of recovery on the affected endpoint.

For intuition, suppose balanced CE attains balanced accuracy 0.72, imbalanced CE attains 0.62, and a mitigation method attains 0.68. Imbalance caused a 0.10 deficit and the method regained 0.06, or 60% of the demonstrated damage. This normalization distinguishes repair of an imbalance effect from a generic method improvement. Exact estimands and gates appear after the endpoints they use.

4.1 Compared methods and shared execution controls

Table 3 lists only the methods evaluated in this benchmark, grouped by the mitigation signal each one reads. CE is the size-matched no-mitigation reference. The roster is fitted in the balanced, moderate, and severe conditions, which alone supply the paired contrast a mitigation claim requires; the natural anchor carries the CE reference only. Method objectives, tested adaptations, ablations, defaults, and fidelity qualifications are defined exclusively in the companion methods report (Queisler, 2026b).

Signal	Tested methods	Intervention
None (reference)	CE	None
Prevalence	Balanced sampling; weighted CE; train-time and post-hoc logit adjustment; cRT	Sampling; loss weight; logits and prior correction; classifier retraining
Nominal support	Class-balanced CE	Loss weight
Independent support	Independent-support weighted CE	Loss weight
Difficulty	Focal loss; pilot-difficulty weighted CE	Loss weight
Diversity	Semantic-scale weighted CE	Loss weight (dynamic)
Nominal support and difficulty	CFAL	Prototype head and margin loss
Prevalence (hybrid)	CE + soft F1; CE + soft MCC; OKO set learning	Sampler and auxiliary loss; set construction and set loss

Table 3: Prespecified patch-regime method roster, grouped by mitigation signal following the companion methods report, with intervention level retained as a second column so the two axes stay visibly orthogonal.

All methods in a controlled comparison retain the frozen encoder, model family, optimizer family, batch definition, augmentation, numerical precision, and optimizer-update budget (Table 5). Hyperparameters and checkpoints are selected only on natural validation data, and are locked before confirmation fitting and the one-time test evaluation. Balanced CE, imbalanced CE, and mitigation runs share split, tail assignment, confirmation-seed index, and held-out cases. Appendix D gives update formulas, adaptive grids, tie-breakers, exposure accounting, failure rules, and cost reporting.

Figure 4 places these steps in execution order. It also separates the validation search that is self-contained from the phase that cannot start until a condition's CE search has resolved, which is what fixes the order in which runs may be launched.

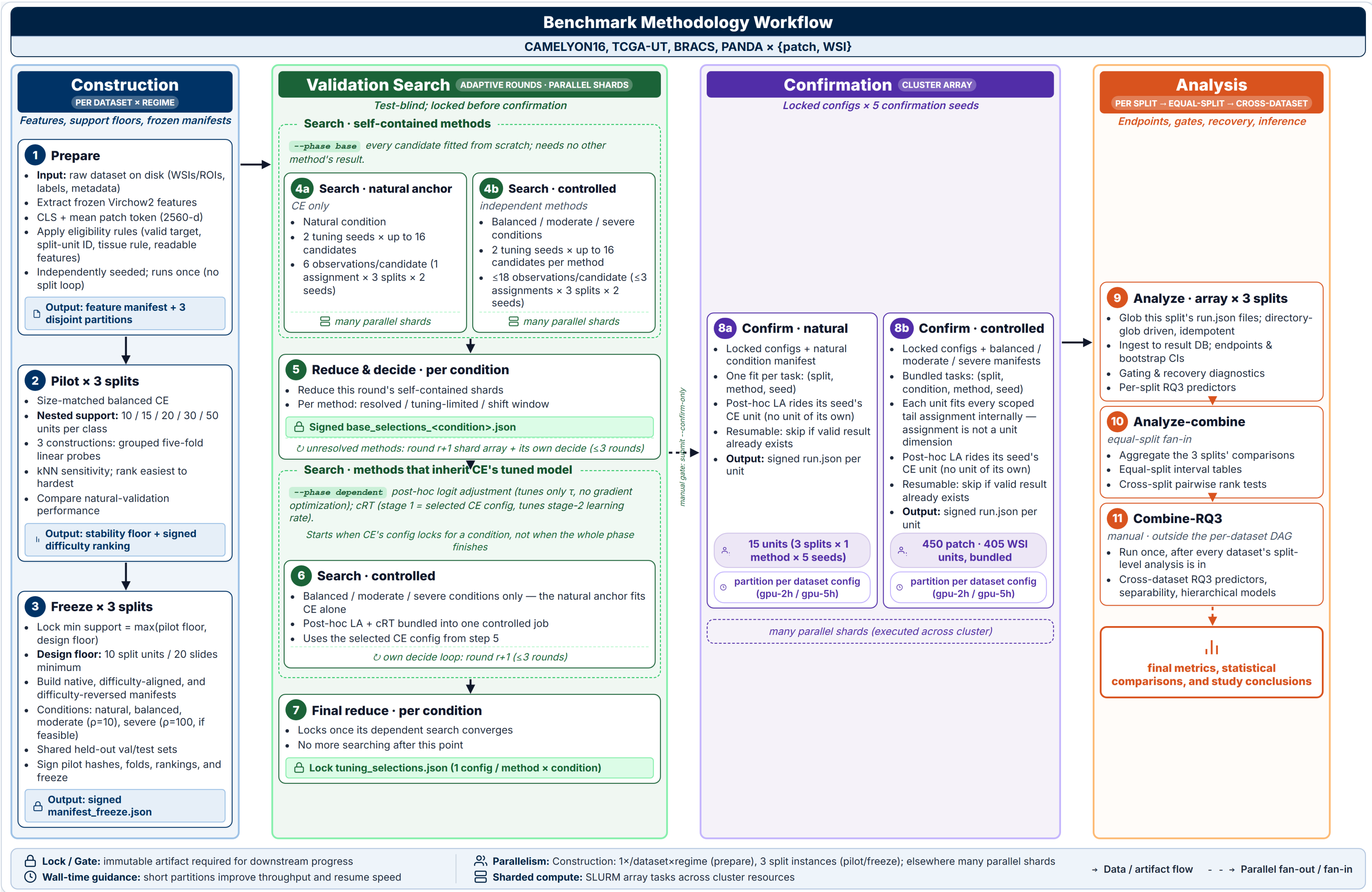

Figure 4: Operational workflow of the controlled class-imbalance benchmark.

This separation makes confirmation runs comparable without pretending that equal optimizer updates imply equal data exposure. With models fixed, evaluation can ask which endpoint was damaged and how much of that damage was recovered.

4.2 Evaluation endpoints

Damage and recovery are evaluated on two complementary axes: discrimination measures whether the predicted class is correct, whereas probability quality measures whether the complete predicted distribution is reliable. Both axes preserve the independent patient or case as the unit of evidence.

Discrimination and aggregation. The primary endpoint is balanced accuracy, equal to macro recall for single-label multiclass prediction. Secondary discrimination outcomes are macro F1; per-class recall and F1; and head, body, and tail recall under the locked assignment. Classes are ranked by allocated training support: the first $\lceil K/3 \rceil$ ranks are head, the last $\lceil K/3 \rceil$ are tail, and intervening ranks are body; binary tasks have one head and one tail and no body. Ties follow the locked class ordering. Overall accuracy is descriptive. Patch predictions are summarized in two ways: patch-micro estimates treat eligible patches as predictions while uncertainty is clustered by partitioning unit, and case-macro estimates first aggregate the metric contribution within each partitioning unit so that heavily tiled cases do not dominate.

Probability quality. Probability-quality metrics ask whether the complete predicted distribution is useful, not only whether its largest entry names the correct class. Tail-group macro negative log likelihood (NLL) is the co-primary calibration endpoint because it gives each tail class equal weight and penalizes confident errors. Because NLL uses the natural logarithm, its values and differences are measured in nats. A nat is the unit of information obtained with logarithms of base e, just as a bit is obtained with logarithms of base 2. Natural-prevalence NLL, macro NLL, multiclass Brier score, expected calibration error (ECE), and reliability diagrams provide secondary views. ECE uses 10 fixed equal-width bins and remains secondary because small classwise samples make bin estimates unstable.

Discrimination uses balanced-decision logits, whereas calibration uses target-prior-corrected probabilities before temperature scaling. Temperature-scaled outputs are secondary and never enter a recovery gate. Appendix B defines every probability-quality equation, clipping at 10^{-12}, score variants, and fixed-bin ECE.

4.3 From imbalance deficit to mitigation recovery

Recovery is meaningful only when imbalance has first caused a measurable loss. The analysis therefore proceeds in two stages: compare size-matched balanced and imbalanced CE, then evaluate mitigation only on an endpoint that shows sufficiently large and statistically supported harm. One *comparison unit* is a fixed dataset, imbalance severity, and tail assignment. A *gate* is the prespecified eligibility rule that routes such a unit to the relevant recovery analysis; it is not a learned model component.

For a higher-is-better metric M, define the imbalance-induced deficit and recovery fraction at each imbalanced condition as

$$\begin{aligned} D_M &= M(\text{balanced CE}) - M(\text{imbalanced CE}), \\ R_M &= \frac{M(\text{method on imbalanced data}) - M(\text{imbalanced CE})}{D_M}. \end{aligned} \tag{3}$$

$R_M = 0$ denotes no recovery, $R_M = 1$ recovery to the size-matched balanced CE reference, $R_M < 0$ further deterioration, and $R_M > 1$ performance beyond that reference. Because the balanced reference draws more unique tail independent units than any imbalanced condition at the same T, methods that only reweight or recalibrate the imbalanced training set cannot restore tail evidence that was never sampled; for that family $R_M < 1$ may reflect an information ceiling rather than method failure, whereas resampling and synthesis methods are the only family for which $R_M = 1$ is mechanically attainable. The full natural condition is descriptive and never enters D_M or R_M: it has no size-matched balanced counterpart, so neither a deficit nor a recovery fraction is defined there. Fitting mitigation methods on it would therefore yield differences confounded by training volume as well as allocation, and the anchor is restricted to its CE reference for that reason.

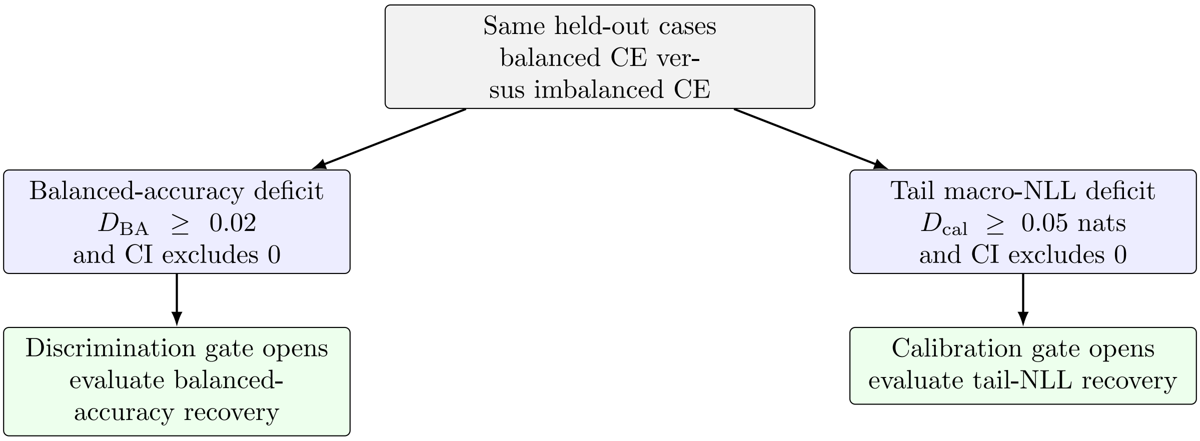

Figure 5: Routing from an imbalance deficit to mitigation analysis. A comparison unit can enter through either gate or both. If neither rule is met, its CE deficits are still reported, but a recovery ratio is not interpreted. CI denotes the paired 95% confidence interval.

Figure 5 shows the two routes. The *discrimination gate* opens when the CE balanced-accuracy deficit is at least 0.02 and its paired 95% confidence interval excludes zero. The *calibration gate* opens when the CE tail-group macro-NLL deficit, $\mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$, is at least 0.05 nats with a paired 95% confidence interval excluding zero. Both rules are computed from CE runs on the same held-out cases before any mitigation method is compared. A comparison unit entering through the discrimination gate is evaluated for balanced-accuracy recovery; one entering through the calibration gate is evaluated for tail-calibration recovery; it may enter through both.

The two routes preserve an important distinction. Strong frozen features may leave balanced accuracy nearly intact while imbalance still biases tail probabilities, in which case a single discrimination rule would exclude methods designed to correct class priors. Conversely, the gates avoid unstable recovery ratios when no meaningful damage exists to recover. All CE deficits on both axes remain reported, including units that fail both rules. For the lower-is-better calibration axis, the sign-corrected deficit $D_{\mathrm{cal}} = \mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$ is the denominator and the method's tail-NLL improvement over imbalanced CE is the numerator, so Equation (3) retains the same interpretation. Recovery for other secondary metrics is descriptive and does not reopen a comparison unit that fails both gates.

4.4 Addressing uncertainty

The deficit and recovery estimates in Section 4.3 summarize a finite held-out cohort rather than fixed population quantities. Their uncertainty has two sources: different patients or cases could produce different estimates, and repeated model fitting introduces algorithmic variation. Multiple patches from the same patient or case do not provide equivalent independent evidence, just as repeated fits do not create additional patient evidence.

Uncertainty is therefore estimated by resampling patients or cases as complete units and repeating the relevant comparisons. The same resampled cohort and corresponding model-initialization draws are used for the balanced CE, imbalanced CE, and mitigation results within each controlled comparison. For intuition, suppose one resample gives a particular patient greater influence. That patient's predictions receive the same greater influence when balanced CE, imbalanced CE, and the mitigation method are evaluated. The deficit and recovery fraction therefore remain comparisons on the same reweighted cohort rather than comparisons between differently sampled groups.

Repeating this process and recomputing the complete deficit and recovery fraction shows how much both estimates vary with cohort composition and model initialization, producing their 95% confidence intervals. Data splits are averaged as design repetitions rather than treated as independent cohorts. Appendix B specifies the resampling construction, validity checks, support diagnostics, and interval calculation.

4.5 Interpreting the evidence

The preceding subsections establish how models are compared, which endpoints record damage, when recovery is interpretable, and how uncertainty is represented. RQ1 is answered by the signed CE deficit and its confidence interval: together they show the direction and magnitude of the allocation effect, while the gate identifies damage large and certain enough to support a recovery analysis. RQ2 begins only for those gate-passing comparisons. A method's improvement over imbalanced CE and its recovery fraction then show how much of the demonstrated damage was repaired. Raw effects and confidence intervals remain the main evidence; recovery fractions are normalized summaries rather than test statistics.

The method roster creates many related method-by-condition comparisons. Treating every result as an independent primary test would increase the chance of a false-positive claim, so the benchmark prespecifies a limited confirmatory family whose tests are adjusted jointly. The confirmatory part of RQ2 asks whether signal choice matters when the intervention mechanism is held fixed. It therefore compares the five weighted-CE methods in Table 3, whose weights use prevalence, nominal support, independent support, difficulty, or diversity while leaving the CE mechanism unchanged. Differences among them can consequently be interpreted as evidence about the signal rather than the intervention. Within each dataset, paired tests support the gate-passing comparisons and Holm adjustment controls their joint family-wise error. Other roster members retain the same effect estimates and confidence intervals but remain exploratory. Appendix B.3 specifies the tests, multiplicity family, and protocol revision; Appendix D.2 gives method-specific diagnostic and tuning details.

Section 4 therefore produces three outputs from each controlled comparison: whether damage occurred, how large it was, and how much of it was recovered when recovery was defined. RQ1 and RQ2 estimate these quantities within a dataset and task. The remaining question is why damage magnitude and recovery vary across datasets, severities, and tail assignments. Section 5 models those two continuous outputs and retains damage occurrence only as the gate that determines when recovery is defined.

5 Explaining Heterogeneity

RQ3 asks which realized signal shortages are associated with damage and recovery across controlled conditions. The unit of analysis is a controlled comparison, not an individual patient. A damage comparison identifies a dataset, severity, and tail assignment; a recovery comparison additionally identifies the method and gate. Damage occurrence is not modelled because the RQ1 gate already provides that thresholded decision.

Outcomes to explain.

1. **Damage magnitude** is the signed balanced-accuracy deficit. Every controlled comparison contributes, including those showing no damage or an apparent improvement under imbalance.

2. **Recovery** is the fraction of the demonstrated deficit repaired by a mitigation method. Only gate-passing comparisons contribute because there is no stable recovery fraction without an established deficit.

Candidate explanations. Predictors are constructed without using test outcomes. Together, the four predictors in Table 4 provide one summary for each distinct dimension of the signal taxonomy.

Predictor	Question it represents	Model role
Achieved allocation severity $\log \rho$	How unequal was the realized head–tail allocation?	Joint model
Independent-evidence shortage S_{ind}	How much patient or slide support did deprived classes lose?	Joint model
Support–difficulty alignment $A = \text{corr}(\log N_c, d_c)$	Did greater support go to harder classes ($A > 0$), or did scarcity compound difficulty ($A < 0$)?	Joint model
Diversity shortage S_{div}	How much representational diversity did deprived classes lose?	Joint model

Table 4: The four RQ3 predictors. Prevalence and nominal support share one allocation-severity predictor because the fixed total T makes them two readings of the same manipulated allocation.

Both shortage measures compare the imbalanced condition with its size-matched balanced condition and include only classes allocated fewer patches under imbalance. Independent-evidence shortage measures the loss of patient or slide support. Diversity shortage measures the loss of semantic-scale log-volume on fixed Virchow2 embeddings, using a within-condition draw matched across classes in both cases and patches. This fixed, pre-mitigation measure reflects the diversity-aware method's dynamic signal without using fitted models or test outcomes.

Exactly two models are fitted: one for signed damage and one for recovery among gate-passing comparisons. Both use the same four standardized predictors and a dataset–target random intercept. Neither includes interactions nor method effects. Weak Gaussian priors and bootstrap observation errors stabilize estimation, while a leave-one-group-out check assesses whether the damage associations depend on one dataset–target group. Appendix C gives the full predictor construction and model form.

The four predictors are correlated summaries of the realized signal profile. Their coefficients therefore describe conditional associations, not separate causal effects. RQ3 is explanatory and hypothesis-generating; it does not claim predictive generalization.

RQ3 can therefore identify signal combinations that deserve follow-up, but the small number of dataset–target groups limits broader inference. The final section places this limitation alongside the benchmark's other interpretation boundaries.

6 Conclusion and Scope

This benchmark is designed to answer a bounded question: when a fixed training budget is redistributed across classes, which signal shortages appear, what damage follows, and which signal-aware interventions recover that damage? Holding the representation, partitions, model family, and optimization budget fixed makes the allocation contrast interpretable. The resulting claims concern changes in the training distribution under one frozen Virchow2 representation.

Several boundaries remain. The design does not establish how end-to-end encoder fine-tuning would respond, whether another foundation representation would produce the same deficits, or whether constructed tails reproduce biological prevalence and referral pathways. The shared total T isolates allocation by withholding some available examples. The full natural condition is therefore retained as a descriptive CE anchor, placing the controlled results alongside the complete training partition and the dataset's natural class prevalence. Because no mitigation method is fitted on that anchor, method comparisons remain confined to the size-matched controlled conditions.

Method comparisons identify methods, not universal mechanisms. Interventions can change sampling, objectives, or architecture together, so equal optimizer updates do not imply equal information exposure. For example, post-hoc logit adjustment performs no gradient optimization, whereas cRT retrains only the classifier in its second stage. Processed examples, unique exposures, accelerator time, memory, and model size are therefore reported rather than assumed to be equivalent.

Evidence also depends on independent support, not merely patch count. BRACS patch classification remains descriptive because its patch-micro Kish effective support falls below the prespecified threshold under patient-level bootstrap weights. Calibration has a separate boundary: it is conditional on the natural held-out prevalence and cannot be transported to another deployment population without an explicit prior-shift analysis.

Within these boundaries, the three research questions form a deliberate sequence. RQ1 establishes the signal profile and damage created by controlled allocation. RQ2 asks which interventions recover performance only after that damage is established. RQ3 then relates variation in damage and recovery to the realized shortages without treating those associations as causal decomposition. Together, this sequence turns class imbalance from a single prevalence label into a controlled analysis of what evidence is missing, what that shortage changes, and which signal a mitigation method should use.

A Construction and Support Specification

This appendix records dataset eligibility, hierarchical sampling, support floors, allocation, and seed rules. These details implement the controlled contrast introduced in Section 3; they do not add further estimands.

Locked quantity	Value	Spec.
Representation	Frozen Virchow2 features	A
Patch classifier	Two-layer MLP, 512 hidden units	A
Update budget	$U = 30\lceil T/B \rceil$ optimizer updates	D
Learning-rate envelope	Seven values, 1×10^{-5} to 1×10^{-2}	D.2
Support floor per class	≥ 10 partitioning units and ≥ 20 slides	A.2
Contribution caps	$\leq 10\%$ per partitioning unit; $\leq 5\%$ per slide	A.2
Severity targets	$\rho_{\text{target}} \in \{1, 10, 100\}$	A
Severity tolerance	Achieved $\rho \in [9, 11]$ and $[90, 110]$	A.2
Tail assignments	Native, difficulty-aligned, difficulty-reversed	A
Data partitions	Three locked, patient- or case-disjoint	A
Initialization seeds	Two tuning; five disjoint confirmation	D

Table 5: Design and training quantities locked before the first definitive fit, with the appendix section that specifies each. Only class allocation within the shared training support T varies across controlled conditions.

A.1 Eligibility and representation

Before partitions are derived, the source manifest is restricted to eligible examples with fixed features. A sample is excluded if it lacks a valid target or partitioning-unit identifier, yields no patch under the dataset-specific tissue rule, lacks an annotation required for its patch target, or lacks readable fixed features. Each remaining partitioning unit is then assigned to exactly one training, validation, or test partition, and all slides from that unit follow its assignment. Eligibility is fixed once rather than reapplied by imbalance condition.

TCGA-UT further excludes one of its 32 annotated cancer types, Diffuse Large B-cell Lymphoma, from the eligible target set. In every locked split this class contributes 20 participants and 20 slides, one slide per participant, so the 10% partitioning-unit and 5% slide patch contribution caps (Section A.2) admit essentially no count above the definitive floor: the class's set of cap-feasible patch counts covers only a few percent of the range between that floor and its native support, with the first infeasible count one unit above the floor. No shared total is therefore both tolerance-feasible and cap-feasible for this class, independent of severity or tail assignment. The remaining 31 cancer types are eligible and satisfy the tolerance search below.

All eligible patches enter the pool before condition-specific subsampling. The common classifier is a two-layer MLP with a 512-unit hidden representation, ReLU activation, dropout, and a linear class output. CFAL replaces the linear output with a prototype-based affinity head; OKO adds a training-only auxiliary head discarded at evaluation. Dataset provenance and complete preprocessing definitions remain in the companion dataset report (Queisler, 2026a).

A.2 Independent evidence and support floors

The independent-support rule of Section 3 is realized as a fixed per-class partitioning-unit pool and, within it, a fixed slide pool, made available to the balanced and all imbalanced conditions: every condition draws its patches only from this pool. Table 6 states the selection order and caps.

The condition with the largest patch allocation for a class covers the fixed pool in full, so no unit in it goes unused by every condition together. A smaller condition for the same class, in

Selection order	Contribution caps	What imbalance changes
Partitioning units, then slides within units, then patches; deterministic round-robin before any unit is revisited	$\leq 10\%$ of a class's patch allocation per partitioning unit; $\leq 5\%$ per slide	Patches per class; the available partitioning-unit and slide pools are unchanged

Table 6: Hierarchical selection and contribution caps. Every sample list reports per-class partitioning-unit, slide, and patch counts, maximum contributions, and the fraction of the eligible independent pool retained.

particular a severe tail count well below that maximum, may leave some pool units untouched; this is the intended shape of an imbalanced allocation, not a violation of the shared pool.

The smallest defensible tail support is estimated before the definitive comparison with a *support pilot*. These balanced CE runs ask when additional independent training units cease to materially improve natural-validation performance; they do not tune a hyperparameter or contribute observations to the definitive analysis. One pilot unit is a partitioning unit, or a slide where partitioning unit and slide coincide. Each selected unit contributes the same fixed quota q_{patch} of patches, distributed as evenly as possible over its eligible slides. The quota is the largest value the patient and slide contribution caps admit at every candidate level and every construction ordering, held fixed so increasing a pilot level adds independent units rather than patches per unit. A caps-admissible quota is necessary but not sufficient: a unit cannot supply more patches than it holds, so a unit is eligible for a candidate level only if its own inventory meets the candidate quota, and the quota at a given level is set from that level's order statistic of per-unit inventory over the eligible units rather than from the scarcest unit in one seeded ordering. A candidate level that no quota can satisfy is dropped from the curve rather than reusing a looser quota from a larger level.

Nested pilots compare $10, 15, 20, 30$, and 50 units per class, or all available units when the scarcest class has fewer. They are repeated for three independently seeded *construction orderings*. An ordering is a fixed sequence of eligible units; its prefix of length m is simply the first m units in that sequence. Thus the 15-unit pilot retains the same first 10 units as the 10-unit pilot and adds the next five. At lower levels of the hierarchy, every partitioning unit has a fixed slide order and every slide has a fixed patch order. Figure 6 illustrates this nesting.

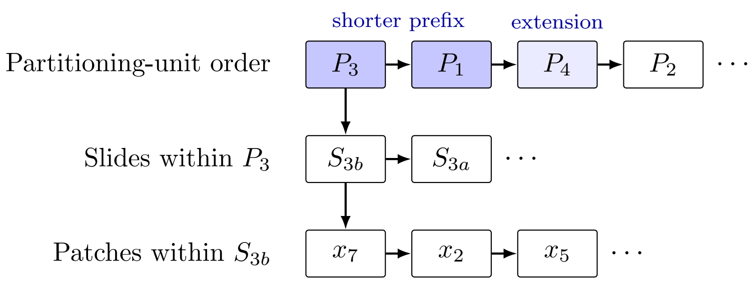

Figure 6: Hierarchy behind one construction ordering. A seed fixes the partitioning-unit sequence and the local slide and patch sequences. The darker unit boxes form a shorter prefix; the lighter third box is added, without replacing earlier units, when the prefix is extended.

Because every larger pilot level extends the preceding prefix, adjacent levels are paired within an ordering and evaluated on the same natural-validation cohort. The smallest level whose next increment changes mean balanced accuracy by less than 0.01 and every class recall by less than 0.02 in all three orderings is the *stability floor*; if none qualifies, the largest candidate is used.[1]

[1] No pilot level is exempt from the caps. A level is dropped from the curve when no quota satisfies both caps for every construction seed—e.g., the 5% slide cap needs at least 20 contributing slides, so a class whose partitioning

The definitive minimum is the larger of this empirical stability floor and a prespecified design floor. Each class must contain at least 10 partitioning units and 20 slides, or 20 slides when partitioning unit and slide coincide, and any class-aware batch or set must contain at least two distinct same-class units. The 10-unit and 20-slide values are not estimated performance thresholds: they are the smallest counts compatible with the 10% partitioning-unit and 5% slide contribution caps, respectively. A requested severity is reduced to the largest allocation satisfying both floors; a dataset is excluded if even its balanced condition fails them. Pilot quotas, curves, seeds, and resulting floors are frozen and reported with the definitive training sets.

Within these floors, one shared total T is fixed for the balanced, moderate, and severe conditions of a split. Construction scores every integer total between the independent-support floor and the largest total any locked tail assignment's head class can support at $\rho_{\text{target}} = 100$, keeping only totals whose balanced allocation and whose every locked assignment's moderate and severe allocation satisfy the patch or slide contribution caps. Among these cap-feasible totals, the *tolerance-feasible* set is those whose achieved moderate ratio falls in $[9, 11]$ and whose achieved severe ratio falls in $[90, 110]$ simultaneously, for every locked assignment. The largest tolerance-feasible total is checked against the true fixed-pool construction — the same sampling procedure the definitive training sets use — starting from the largest candidate and moving to the next largest until one is confirmed; that confirmed total is T. A dataset–regime with no tolerance-feasible total retains its largest independent-support- and cap-feasible total instead, subject to the infeasibility fallback and degeneracy rule below.

Before assigning support ranks, the largest balanced pilot level is used to measure class difficulty. Each of the three pilot constructions supplies one hierarchically matched draw in the sense of the companion methods report (Queisler, 2026b): every class contributes the same number of partitioning units and every selected unit the same number of patches, so the probe sees equal independent support in every class rather than only equal patch counts. On each draw a class-balanced linear probe makes five-fold out-of-fold predictions with case/patient-disjoint `StratifiedGroupKFold` partitions. Class difficulty is $d_c = 1 - \overline{\text{recall}}_c$, where recall is averaged across the three matched draws; five-neighbour kNN recall and the unmatched, patch-balanced-only probe error are retained as sensitivity evidence, the latter because a class ordering that changes between the matched and unmatched scores marks a class whose apparent difficulty is partly scarcity. Pilot hashes, folds, classwise scores, and the easiest-to-hardest ranking are signed. Equal scores retain native class order.

A.3 Severity profiles and tail assignments

A binary task needs only a head and a tail count, but a multiclass task also needs a rule for the classes between those extremes. The exponential construction places the K class supports at equal intervals on a logarithmic scale. For a locked order from head ($i = 0$) to tail ($i = K - 1$), it assigns weights

$$w_i(\rho_{\text{target}}) = \rho_{\text{target}}^{-i/(K-1)}, \qquad n_i^* = T \frac{w_i(\rho_{\text{target}})}{\sum_{j=0}^{K-1} w_j(\rho_{\text{target}})}, \quad i = 0, \ldots, K-1. \tag{4}$$

so the head-to-tail ratio is exactly ρ_{target} before integer rounding. For example, with four classes and $\rho_{\text{target}} = 100$, the unnormalized weights are approximately $1, 0.215, 0.046$, and 0.01; the two body classes lie geometrically between the head and tail instead of being assigned ad hoc counts. Binary targets use $n_{\text{head}}^*/n_{\text{tail}}^* = \rho_{\text{target}}$ directly. The same exponential rule is applied to every multiclass dataset–regime, so a given ρ_{target} denotes the same allocation shape in each of them.

units mostly contribute a single slide can leave levels below 20 infeasible. Dropped levels are recorded in the frozen pilot report.

The values n_i^* are real-valued targets, whereas a training set must contain whole patches or slides. *Constrained integer allocation* converts them into usable class counts. Each target is rounded to the nearest integer and clipped to the class's support floor and available unique support. If the rounded counts do not sum to T, units are added to or removed from the classes with the largest rounding residuals until the total is exact. The resulting integers n_i therefore satisfy $\sum_i n_i = T$ and every class-specific lower and upper bound while remaining close to the exponential targets under this deterministic adjustment.

Counts determine how many units a class contributes; construction orderings determine which units they are. One master ordering is created per class at each hierarchy level. Every condition selects a deterministic prefix, or a partitioning-unit-round-robin extension of a prefix, from these same orderings: all fixed-pool partitioning units and slides contribute before additional patches are allocated.

If a target ratio is infeasible, the same rule is applied to every dataset and regime: use the largest attainable ratio and report the requested ratio, achieved ratio, limiting class, realized counts, and binding support floor. Validation or test support is never reduced to manufacture feasibility. This general fallback is especially relevant to high-K tasks, where an exponential profile must allocate support to many intermediate classes as well as the tail; it is not a dataset-specific exception. A stable lower-severity condition is preferred to a nominal $\rho = 100$ condition supported by only a few independent units.

The fallback has a limit: a moderate or severe condition whose realized allocation is indistinguishable from the balanced condition—achieved ratio within a small tolerance of 1, or realized counts identical to the balanced training set—is not a lower-severity instance of the intended contrast but a null one, since the imbalance intervention did not occur. Such a condition is reported as degenerate rather than fitted and averaged with genuine moderate or severe conditions from other splits or assignments, and it is excluded from the imbalance deficit and recovery estimands in Section 4.3. Its requested ratio, achieved ratio, limiting class, and binding support floor are recorded alongside the exclusion, using the same fields every condition already reports.

The profile determines support by rank; the tail assignment maps semantic classes to those ranks. Native order is the clinical, ordinal, or natural-support order of the target; aligned and reversed order the classes by the difficulty scores d_c measured above. Binary tasks retain native and reversed orientations only. A difficulty mapping that duplicates an earlier mapping in any locked split is omitted across that dataset–regime with its reason recorded. These three assignments separate interpretable mechanisms but do not exhaust all possible class-support mappings. Figure 3 illustrates the design.

The random quantities are separated into five seed families so that changing one source of variation does not silently change another. Table 7 defines their roles.

Seed family	Determines	Prespecified use
Data partition	Membership of the training, validation, and test cohorts	Three locked patient- or case-disjoint splits
Tail assignment	Mapping of semantic classes to head–tail support ranks	Native, difficulty-aligned, and difficulty-reversed mappings from signed pilot evidence
Construction	Partitioning-unit, slide, and patch orderings and the selected evidence	Three pilot orderings and a disjoint fourth ordering for definitive training sets
Initialization	Parameter initialization and optimization randomness	Two tuning seeds and five disjoint confirmation seeds
Resampling	Bootstrap draws and permutation swaps	Uncertainty intervals and hypothesis tests only

Table 7: Seed families and their non-overlapping roles in the benchmark.

No seed is chosen based on test performance. The size-matched balanced training set and its trained CE baseline are shared by all tail assignments within a split.[2] The same balanced predictions enter every assignment-specific deficit and are represented once in resampling data, avoiding duplicated balanced fits, bootstrap records, and artificial assignment variability.

B Endpoint and Inference Specification

This appendix defines secondary endpoints, score semantics, resampling, diagnostics, and confirmatory testing. Gates and recovery retain the main-text definitions in Section 4.3.

B.1 Probability quality and calibration

Let N denote the number of held-out predictions in an aggregation unit, C the number of classes, $y_i \in \{1, \ldots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0, 1]^C$ the predicted probability vector. Write Y_i for the one-hot encoding of y_i.

Table 8 fixes which score variant enters each endpoint. For ordinary CE, post-hoc logit adjustment, and train-time logit adjustment, the target-prior variant is the correction defined in the companion methods report, using the natural-validation prior for the corresponding test split. For every other method, the target-prior variant equals its raw logits.

Endpoint or output	Score variant
Balanced accuracy and all other discrimination endpoints	Balanced-decision logits: $z^{\mathrm{CE}} - \tau \log \widehat{\pi}$ for post-hoc logit adjustment; raw model logits for train-time logit adjustment and every other method.
Tail-NLL deficit gate and tail-NLL recovery	Target-prior-corrected probabilities before temperature scaling.
Reported NLL, Brier score, ECE, and reliability diagrams	Target-prior-corrected probabilities before temperature scaling; raw probabilities are retained as a separately labelled diagnostic.
Temperature-scaled calibration outputs	Target-prior-corrected logits divided by one positive temperature fitted on natural-validation NLL and applied unchanged to test; these secondary outputs do not enter either recovery gate.

Table 8: Canonical mapping from evaluation endpoints to score variants.

Negative log likelihood. Natural-prevalence negative log likelihood (NLL), also called multiclass log loss, is

$$\mathrm{NLL} = -\frac{1}{N} \sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{5}$$

Lower NLL indicates that more probability mass is assigned to the true class. The implementation clips $\hat{p}_{i,y_i}$ to $[10^{-12}, 1]$ before taking the logarithm. To prevent common classes from dominating probability-quality summaries, class-conditional NLL and macro NLL are

$$\mathrm{NLL}_c = -\frac{1}{N_c} \sum_{i:y_i=c} \log \hat{p}_{ic}, \qquad \mathrm{NLL}_{\mathrm{macro}} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{NLL}_c, \tag{6}$$

where N_c is the number of held-out predictions from class c. Natural-prevalence NLL is the deployment-facing summary for the observed held-out cohort; macro NLL and unweighted head, body, and tail averages of NLL_c are tail-sensitive summaries.

[2] At $\rho_{\mathrm{target}} = 1$, every class receives equal support. Permuting semantic classes across support ranks therefore leaves the allocation unchanged, so tail assignment is not a meaningful factor for the balanced condition.

Multiclass Brier score. The multiclass Brier score penalizes squared error across the full predicted distribution (Brier, 1950):

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2 \,. \tag{7}$$

Lower values indicate better probability quality. The natural-prevalence score averages this error over the complete held-out cohort. Classwise scores apply the same full-distribution error only to predictions with $y_i = c$. Their unweighted averages provide head, body, and tail summaries.

Expected calibration error. For each prediction, let $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and $q_i = \max_c \hat{p}_{ic}$. Expected calibration error (ECE) partitions confidence into $B = 10$ fixed equal-width bins on $[0, 1]$ and compares mean confidence with empirical accuracy in each nonempty bin (Naeini et al., 2015):

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left| \frac{1}{|I_b|}\sum_{i\in I_b} \mathbb{1}[\hat{y}_i = y_i] - \frac{1}{|I_b|}\sum_{i\in I_b} q_i \right| , \tag{8}$$

where I_b contains the predictions assigned to bin b. ECE is secondary because it depends on binning and can be unstable with small classwise samples; its bootstrap uncertainty uses the same fixed bins and partitioning-unit weights as the other endpoints. Reliability diagrams show the binwise confidence–accuracy relation overall and for the tail group.

Because checkpoints and hyperparameters are selected for natural-validation balanced accuracy, all probability-quality results describe discrimination-selected models rather than models optimized for calibration. The output semantics and prior-correction formulas of individual methods are defined in the companion methods report (Queisler, 2026b).

B.2 Uncertainty from resampling partitioning units

The resampled partitioning unit is the patient or participant where one is available, and the biopsy/WSI for PANDA and CAMELYON16. Each replicate reweights the observed partitioning units rather than deterministically repeating each one exactly once, and the complete statistic is recomputed from those weights.

Label-only preflight. Before model training, 10,000 bootstrap replicates are constructed using only partitioning-unit identifiers, split membership, and class labels. *Label-only* means that no predictions, performance values, or method identities are used; these replicates test whether the planned resampling scheme is valid before results can influence it. The count of 10,000 is a prespecified Monte Carlo budget. It places about 250 replicates in each 2.5% tail of a percentile interval, reducing simulation noise in the interval endpoints while remaining inexpensive at this label-only stage.

Each partitioning unit draws an independent, continuous bootstrap weight in every replicate: a Bayesian bootstrap (Rubin, 1981) in which the n observed units' weights are n times a draw from a symmetric Dirichlet$(1, \ldots, 1)$ distribution over the n-simplex. This draw is almost surely strictly positive in every coordinate, so every unit contributes to every replicate; no split–class cell, including one contributed by a unit whose observed split-by-class pattern is unique among the cohort, can be emptied by an unlucky draw. A unit's observed contribution pattern—how many examples it supplies to each class in each test-split appearance—is still recorded and grouped into strata, but only to report per-stratum diagnostics; it no longer constrains the weight draw itself. A selected unit's weight is applied to all of its slides and patches in every split. Figure 7 summarizes the workflow.

The preflight verifies that every original split–class cell remains represented, every split-level metric is computable, one partitioning unit receives the same weight across all of its split

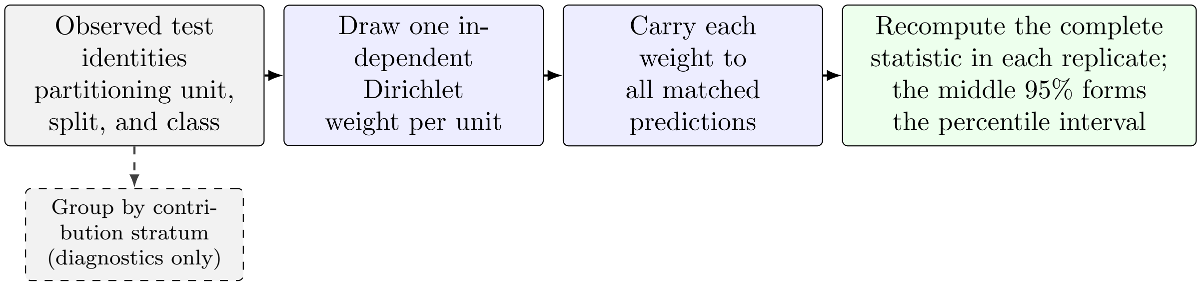

Figure 7: Bayesian bootstrap workflow clustered by partitioning unit. The label-only preflight executes the identity, weight-draw, and stratum-diagnostic steps before predictions exist. After model fitting, the same paired per-unit weights are applied to every matched run.

appearances, and every unit's resampled weight actually varies across replicates. Failure of any check stops the analysis rather than causing a replicate to be silently discarded.

The independent per-unit weight draw can still produce a replicate dominated by a few heavily weighted units. If partitioning unit i has class-specific bootstrap weight v_i, the *Kish effective partitioning-unit count*

$$n_{\mathrm{Kish}} = \frac{\left(\sum_i v_i\right)^2}{\sum_i v_i^2}. \tag{9}$$

translates unequal case weights into the equivalent number of equally weighted partitioning units. It decreases as weight concentrates. The preflight also records each split/class cell's number of contributing units and its largest unit-weight share. Table 9 lists the three concentration checks; failing any one designates inference for that dataset–regime descriptive before training.

Diagnostic	Threshold for descriptive-only designation
Effective partitioning-unit count	2.5th percentile of n_{Kish} across replicates below five
Single-unit dominance	One unit supplies more than 50% of a class's weight in more than 5% of replicates
Cell support	A split/class cell has fewer than five contributing units

Table 9: Label-only preflight concentration checks. All three are evaluated before predictions exist; the designation is recorded rather than used to discard replicates.

A percentile rather than the minimum is used because the minimum across replicates is an extreme order statistic: its value falls as the replicate count rises, so a floor placed on it would encode the resampling budget rather than the support the cell actually carries. The diagnostic report and resampling seed are frozen with the analysis plan.

Intervals after model fitting. The three partitioning-unit-disjoint data splits are fixed design repetitions and are averaged with equal weight; they are not treated as three independent cohorts. Tail assignments remain separate unless an assignment-averaged estimand is explicitly reported. Within each case-bootstrap replicate, the five matched confirmation-seed indices are also resampled as paired blocks and averaged before the splits are combined. This captures algorithmic variation without pretending that five model initializations create five additional case cohorts.

Balanced CE, imbalanced CE, and each mitigation method receive the same partitioning-unit multiplicities and confirmation-seed draws. The complete deficit and recovery fraction, including numerator and denominator, is recomputed in each replicate. The reported 95% percentile interval spans the 2.5th to 97.5th percentiles of the resulting distribution. Patches are never bootstrapped independently.

B.3 Paired testing and multiplicity control

Hypothesis tests provide a supporting check for the prespecified primary methods. On the discrimination axis, the tested effect is the balanced-accuracy improvement over imbalanced CE. On the calibration axis, it is the reduction in tail-group macro NLL, oriented so that positive values indicate improvement. Recovery fractions are normalized effect summaries, not test statistics.

The raw two-sided p-value comes from a paired partitioning-unit-block permutation test. This test represents the null hypothesis of no method difference by repeatedly swapping the method and imbalanced-CE labels within a partitioning unit. The same swap is used for all of that unit's patches, split appearances, and corresponding confirmation-seed indices, preserving the paired design. All possible swaps are enumerated when feasible. Otherwise, 100,000 random swaps are used with a plus-one correction; this gives a minimum attainable Monte Carlo p-value of approximately 10^{-5} and avoids reporting a simulated value of zero.

A *test family* is the set of related hypotheses interpreted together. Testing many hypotheses increases the chance that at least one small p-value occurs by chance. Holm adjustment controls this family-wise error by ordering the raw p-values from smallest to largest, applying the strongest correction to the smallest value, and then proceeding stepwise with less stringent corrections. It provides this control without assuming that the tests are independent.

The confirmatory family holds one method per signal: weighted CE (prevalence), class-balanced CE (nominal support), independent-support weighted CE (independent support), pilot-difficulty weighted CE (difficulty), and semantic-scale weighted CE (diversity). All five share one intervention level, a mean-one class weight on unmodified CE (Section 4.5), so a difference among them is attributable to the signal rather than the mechanism. Within one dataset, the family includes every gate-passing combination of primary method, gate, severity, and tail assignment; severities are not split into separate families. Holm adjustment is applied across the testable comparisons, while gated-out comparisons are recorded as not tested. Every other roster member remains exploratory and outside the confirmatory correction.

This family replaces the intervention-based family originally specified here, whose rationale assumed a second prediction regime that the benchmark no longer carries. The change was fixed before any recovery outcome was inspected. The signal-based family is confirmatory only under that timing and must not be revised again after recovery outcomes are read.

C RQ3 Model Specification

This appendix fixes the four signal predictors and two association models used for RQ3. No predictor uses test labels, deficit estimates, recovery estimates, or fitted mitigation-model outputs.

Balanced-to-imbalanced shortages. For controlled comparison j, let $\mathcal{T}_j = \{c : N_{cj}^{(\text{imb})} < N_c^{(\text{bal})}\}$ contain the semantic classes receiving fewer nominal patches than in the shared balanced condition. The independent-evidence shortage is

$$S_{\text{ind},j} = \frac{1}{|\mathcal{T}_j|} \sum_{c \in \mathcal{T}_j} \log \frac{G_c^{(\text{bal})}}{G_{cj}^{(\text{imb})}}, \tag{10}$$

where G_c counts contributing patients or slides. Positive values mean that nominally deprived classes also lost independent evidence.

Representational diversity is measured on frozen Virchow2 features using the same within-class semantic volume as semantic-scale weighted CE,

$$S'_c = \tfrac{1}{2} \log_2 \det\left(I + \frac{d}{M\varepsilon^2} Z_c Z_c^\top\right). \tag{11}$$

Within each condition, the deterministic draw contains the same number of patients and patches per class, and pooled isotropic scaling fixes $\varepsilon = 1$. With the numerical floor $\varepsilon_S = 10^{-3}$, diversity shortage is

$$S_{\mathrm{div},j} = \frac{1}{|\mathcal{T}_j|} \sum_{c \in \mathcal{T}_j} \log \frac{\max(S_c'^{(\mathrm{bal})}, \varepsilon_S)}{\max(S_{cj}'^{(\mathrm{imb})}, \varepsilon_S)}. \tag{12}$$

Positive values indicate lower observed feature volume among nominally deprived classes. If no class is deprived, both shortage contrasts are defined as zero.

Two association models. For either signed balanced-accuracy damage or gated recovery, the linear predictor is

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\mathrm{ind}} S_{\mathrm{ind},j} + \beta_A A_j + \beta_{\mathrm{div}} S_{\mathrm{div},j}, \tag{13}$$

where $u_{g[j]}$ is the random intercept for the dataset–target group and $A_j = \mathrm{corr}(\log N_{cj}, d_c)$. Every predictor is standardized before fitting. The damage model includes every controlled comparison and incorporates its bootstrap standard error; the recovery model includes only comparisons passing a prespecified RQ1 deficit gate and incorporates the recovery-ratio bootstrap standard error. No occurrence model, interaction, method effect, alternative predictor model, or diagnostic model is fitted.

Parameters are estimated by maximum a posteriori (MAP) fitting with zero-centered unit-scale Gaussian penalties on standardized slopes and log-scale parameters. A leave-one-dataset–target-group-out check refits the damage model while withholding one complete group at a time. This is only a coarse stability check, not evidence of predictive generalization. Model definitions, predictor roles, and priors are frozen before recovery outcomes are inspected, and all RQ3 conclusions remain hypothesis-generating.

D Training, Tuning, and Execution Specification

This appendix records exact update budgets, validation-search rules, replication, and resource accounting. Main-text interpretation depends only on their matched and test-blind use.

D.1 Update budgets and checkpoint selection

The frozen encoder, classifier family, optimizer family, batch definition, augmentation, and numerical precision are held fixed. Training is budgeted only in optimizer updates; epochs are neither a stopping rule nor a unit of comparison. With B the locked batch size and T the common controlled support, the reference budget is

$$U = 30 \left\lceil \frac{T}{B} \right\rceil \tag{14}$$

updates, corresponding to 30 reference passes through T examples under ordinary sampling. Table 10 allocates that budget to the methods that do not consume it in a single stage.

Method	Optimizer updates
Single-stage default	U; every method not listed below
cRT	U stage one, then $\lceil 0.2U \rceil$ stage two on the classifier only
Natural anchor (CE only)	Equation (14) at its own support; descriptive, enters no matched contrast

Table 10: Allocation of the reference update budget U. Balanced sampling and replacement do not change U. Processed examples, unique-example exposures, and updates are all reported.

Within a run, the reported checkpoint is fixed by three rules:

1. Validation metrics are stored at fixed update intervals. The interval is scaled per dataset so the number of stored validation checkpoints is comparable across the widely differing controlled supports T, and the resolved value is recorded with every run.
2. The selected checkpoint maximizes natural-validation balanced accuracy; macro F1 and then lower NLL are deterministic tie-breakers.
3. Test predictions are produced once from that locked checkpoint.

D.2 Control grids and adaptive validation search

Hyperparameter comparisons are bounded by a common validation-search budget rather than by restricting every method to one tuned quantity, and the search adapts within a prespecified envelope rather than committing to a single fixed grid. Every trainable method draws its learning rate from the common envelope

$$\eta \in \{1 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}, 1 \times 10^{-2}\}. \tag{15}$$

Table 11 lists each method's imbalance-specific control. Focal loss, CE + soft F1, and CE + soft MCC draw their strength from a seven-value envelope searched through the same adaptive window as the learning rate; every other control is a fixed four-value grid, since its domain is naturally bounded. Both axes follow the search below, in which an edge winner is never accepted as an optimum: a bounded number of window shifts replaces a single fixed grid, at the cost of one new candidate per shift.

Matched counting-unit diagnostic. Class-balanced and independent-support weighted CE both weight by $1/E_c(\beta)$, the inverse effective number from the same saturating transform, evaluated at nominal count n_c for the former and independent count G_c for the latter, as defined in the companion methods report (Queisler, 2026b). Because β sets both a saturation scale and, jointly with the counting unit, the realized weight shape, tuning β separately for each method confounds the counting unit with the selected saturation scale. The tuning grid therefore includes a *matched-beta* arm: independent-support weighted CE is confirmed a second time at class-balanced CE's tuned β, with its own tuned learning rate. Only this matched-β comparison attributes a difference to the counting unit alone; it is a diagnostic arm reported alongside, not in place of, the five-family contrast, and is not itself one of the five confirmatory members.

Adaptive window search, run once per method, dataset, and imbalance severity.

1. Initialize each active window to four adjacent envelope values: $\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}\}$ for the learning rate, and the start listed in Table 11 for an adaptive strength axis.
2. Fit the active learning-rate window crossed with the method's strength window or fixed grid: at most 16 candidates in any one round. Candidates fitted in an earlier round are reused, never retrained.
3. If an axis's winning value is interior to its window, record the axis as *resolved* and stop searching it.
4. If the winner is an edge value and the envelope extends further in that direction, shift the window one position toward that edge and return to step 2. Only the single newly added candidate is fitted; the three overlapping candidates are reused.
5. If the winner is an edge value at the envelope's boundary, retain the boundary value and record the axis as *tuning-limited*.

Round bounds. At most three rounds on the learning-rate axis, three on focal loss's γ axis, and two on the soft-F1 and soft-MCC strength axis, each bounded by its envelope's length. A resolved or tuning-limited axis is never retrained in a later round.

Method or family	Candidate method-specific control
Balanced sampling / weighted CE	strength $\in \{0.25, 0.5, 0.75, 1\}$
Focal loss	$\gamma \in \{0, 0.5, 1, 1.5, 2, 4, 8\}$ envelope; active window starts at $\{0.5, 1, 1.5, 2\}$
Logit adjustment	$\tau \in \{0.25, 0.5, 1, 2\}$
CE + soft F1 / soft MCC	auxiliary weight $\in \{0, 0.25, 1, 4, 16, 64\}$ envelope; active window starts at $\{0.25, 1, 4, 16\}$
CFAL	affinity bandwidth $\sigma \in \{0.25, 1, 2, 4\}$
OKO	odd-class count $k \in \{1, 2, 4, 8\}$, capped by $K - 1$
Class-balanced CE	$\beta \in \{0.9, 0.99, 0.999, 0.9999\}$; saturation points $1/(1 - \beta) \in \{10, 100, 1000, 10000\}$, matched to n_c's range
Independent-support weighted CE	$\beta \in \{0.8, 0.95, 0.99, 0.999\}$; saturation points $\{5, 20, 100, 1000\}$, scaled down to G_c's range, plus a matched-β arm at class-balanced CE's tuned β
Pilot-difficulty weighted CE	$\tau \in \{0.25, 0.5, 0.75, 1\}$
Semantic-scale weighted CE	$\tau \in \{0.25, 0.5, 0.75, 1\}$; reweighting begins at reference pass 6 of 30

Table 11: Prespecified method-specific controls, crossed with the active window of the learning-rate envelope in Equation (15). Focal loss and the two soft-label hybrids search their listed envelope through an adaptive four-value window; every other row is a fixed four-value grid. cRT has no imbalance-strength grid; only its stage-two learning rate is selected separately. The class-balanced and independent-support β grids are matched saturation points on the effective-number transform, scaled to each method's own counting unit as defined in the companion methods report (Queisler, 2026b).

For weighted CE, balanced sampling, focal loss, CE + soft F1, and CE + soft MCC, an imbalance-specific strength of zero is mathematically identical to plain CE at the same learning rate: it applies no class reweighting or sampling adjustment, reduces the focal term to the plain cross-entropy loss, or drops the auxiliary term entirely. That value is therefore excluded from the searched window, and its metrics are copied from CE's already-fitted same-learning-rate run rather than retrained, giving each of these five methods one additional comparison point at no extra training cost.

Four method families deviate from the search above. CE and methods without a meaningful imbalance-specific control search the learning-rate envelope alone. Train-time logit adjustment is tuned jointly over τ and learning rate, with τ held to its fixed grid. Post-hoc logit adjustment inherits the selected CE model and tunes only τ, because it performs no gradient optimization. cRT inherits the selected CE configuration for stage one and selects its stage-two classifier learning rate separately from the common grid. Temperature scaling, when reported, fits one positive scalar on natural-validation NLL after model and checkpoint selection and applies it unchanged to test logits (Guo et al., 2017).

D.3 Replication, locking, and records

Selection and replication follow prespecified units rather than per-run judgement:

- **Tuning seeds.** Every candidate is fitted with the same two tuning initialization seeds for each prespecified data split and tail assignment.
- **Selection unit.** One method, dataset, and imbalance severity. Within it, natural-validation balanced accuracy is averaged across both tuning seeds, all tuning splits, and all tail assignments; the highest aggregate is selected, with macro F1 and then lower NLL as deterministic tie-breakers.
- **Locking.** The chosen configuration is locked before any of the five disjoint confirmation initialization seeds is fitted and before test evaluation. Tuning runs never enter confirmation effect estimates.

- **Confirmation.** Each locked data split and tail assignment is repeated with the five confirmation seeds. Exactly three partitioning-unit-disjoint split seeds are used; a dataset–regime unable to produce all three valid splits under the independent-support floors is excluded from the corresponding confirmatory analysis.
- **Matching.** The two runs of a controlled contrast (Appendix E) share data split, tail assignment, confirmation-seed index, and held-out cases, so the contrast does not mix the intended change with a different cohort or replication.
- **Failures.** The unit of replication and all missing or failed runs are reported. A failed run is never silently replaced using a test-informed seed.

Split counts, support floors, condition training sets, tuning and confirmation seeds, method grids, and update budgets are frozen in the signed analysis plan before the first definitive fit. Every candidate fitted, every window shift, and the resolved-or-tuning-limited outcome of each axis are recorded there before the corresponding method enters confirmation. No test data and no confirmation-stage result ever informs a shift, and no grid is changed once confirmation training begins.

For every realized condition, the experiment record contains

- dataset version and eligibility rules;
- partitioning-unit, slide, and patch identifiers by partition;
- requested and achieved T and ρ, and the class assignment;
- all seed families and their tuning, confirmation, pilot, or definitive roles;
- pilot patch quota and bootstrap-preflight diagnostics;
- model and optimizer configuration, candidate grid, and selected checkpoint;
- environment identifier and test-prediction hash.

Deviations are dated and justified before the affected test labels are accessed.

Computational cost is separated into validation search and locked-configuration fitting, and reported as wall-clock training time, accelerator-hours, and peak accelerator memory; total and trainable parameter counts; and effective passes through unique training examples. Costs include every method-specific stage, including both cRT stages, but exclude the one-time frozen-feature extraction shared by all methods. Hardware and software environments are recorded with each run.

E Terminology

Natural anchor. The full eligible natural training partition, fitted with the CE reference alone. It is descriptive and may have a different size from controlled conditions.

Size-matched balanced reference. A training subset with total support T and approximately equal class counts; it defines the counterfactual reference in D_M.

Imbalance severity. The requested training allocation summarized by ρ; achieved severity is computed from realized integer counts.

Pilot level. A candidate number of independent training units per class used in the balanced CE support-stability study.

Tail assignment. The locked mapping from semantic classes to positions in the constructed support profile.

Comparison unit. One dataset, imbalance severity, and tail assignment evaluated together.

Deficit gate. A prespecified threshold rule that determines whether an observed discrimination or calibration deficit is large and precise enough for recovery analysis.

Imbalance deficit. The paired balanced-CE minus imbalanced-CE difference at common T and on the same held-out cases.

Recovery. A method's improvement over imbalanced CE divided by the evidenced imbalance deficit.

Kish effective partitioning-unit count. The equally weighted partitioning-unit count with the same weight concentration as one bootstrap replicate.

Intraclass correlation. The estimated similarity of class margins from patches belonging to the same partitioning-unit or slide cluster.

Patch-micro estimate. A metric computed over eligible patch predictions, with uncertainty clustered by partitioning unit.

Case-macro estimate. An equal-weight aggregation over slides or partitioning units, limiting domination by heavily tiled cases.

Self-contained search. The validation-search phase in which every candidate is fitted from scratch, needing no other method's result.

CE-inherited search. The validation-search phase for post-hoc logit adjustment and cRT, which reuse a condition's selected CE configuration and tune only what remains open for that method (τ for post-hoc logit adjustment; the stage-two learning rate for cRT), and so cannot begin until that condition's CE search has resolved.

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